TITLE: **Box–Jenkins in practice: forecasting CO_2 at Mauna Loa, and what an ARMA model finds in the gold price**

SUBJECT: Aim, method and results of version 2 of the `ARMA_CO2_gold` repository, an application of an undergraduate time-series course

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Key finding. The forecasts of version 1 fail for a reason the course explains: a series with a trend and a twelve-month cycle needs a seasonal model with both differences. The seasonal ARIMA $(0,1,1)(0,1,1)_{12}$, chosen by AIC among 36 candidates, forecasts the 48 months of 1998–2001 with a mean absolute error of 0.78 ppm, against 2.30 and 3.80 ppm for the two models of version 1. For gold, monthly returns show no autocorrelation an ARMA model could use: **the historical mean is the forecast to beat**, and neither an ARMA model nor the random walk beats it.

1. Data and design

- **CO_2.** Weekly flask samples at Mauna Loa, 1958-03 to 2001-12, in parts per million (ppm), as shipped with `statsmodels`.[1] The monthly value is the mean of the month's samples, as in version 1. Five months have no sample (1958-06, 1958-10, 1964-02, 1964-03, 1964-04); version 1 filled them with the next month, version 2 starts in January 1965 and fills nothing.
- **Split (version 1).** Estimation to December 1997 (396 months in version 2); one forecast, 1 to 48 months ahead, for January 1998 to December 2001.
- **Gold.** Closing price on Borsa Istanbul, US dollars per troy ounce, last close of each month, 2018-07 to 2026-09 (99 months), from the CBRT data system EVDS (series TP.ALTINPIYASA.KAP03), the earliest month it offers. Version 1's file (from March 2012) is not in its repository.

2. Identifying the CO_2 model

Model. With the backshift operator $By_t = y_{t-1}$, a seasonal ARIMA$(p,d,q)(P,D,Q)_{12}$ is

$$\phi_p(B)\,\Phi_P(B^{12})\,(1-B)^d(1-B^{12})^D y_t = \theta_q(B)\,\Theta_Q(B^{12})\,\varepsilon_t, \qquad (1)$$

[1] Keeling, C. D., Whorf, T. P. (2004). Atmospheric CO_2 concentrations derived from flask air samples at sites in the SIO network. In *Trends: A Compendium of Data on Global Change*. CDIAC, Oak Ridge National Laboratory.

with ε_t white noise, estimated by exact Gaussian maximum likelihood. Version 1 used $P = D = Q = 0$.

The series (Figure 1). CO_2 rose from 316.1 to 371.0 ppm. The change over twelve months averaged 1.09 ppm in 1965–74 and 1.48 ppm in 1988–97, and reached 2.68 ppm in 1998, the first test year. Around the trend, a cycle of 6.2 ppm peaks in May (+2.98) and bottoms out in October (−3.21).

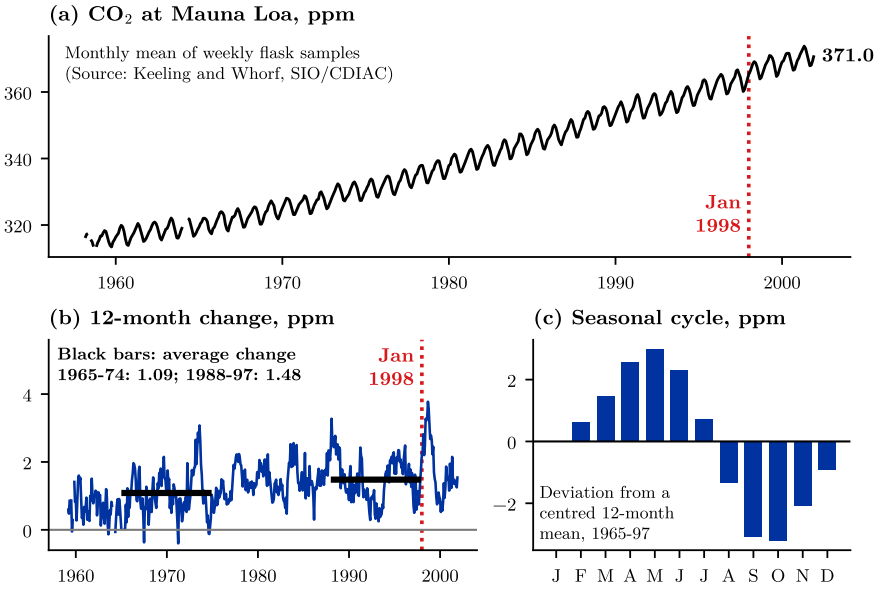


Figure 1. CO_2 at Mauna Loa. (a) Monthly level; the red line marks the first test month. (b) Change over twelve months, $y_t - y_{t-12}$. (c) Seasonal effects: deviations from a centred 2×12 moving average.

Differencing. The ADF test estimates $\Delta x_t = c + \gamma x_{t-1} + \sum_{i=1}^{k} a_i \Delta x_{t-i} + u_t$ and tests H_0: $\gamma = 0$ (unit root); KPSS tests the opposite null, stationarity. On January 1965 to December 1997 the level has a unit root (ADF $p = 0.27$; KPSS $p \leq 0.01$). After one difference both tests point to stationarity (ADF $p < 0.001$; KPSS $p \geq 0.10$), but neither looks for a seasonal unit root; the correlogram does (Figure 2). The autocorrelations of Δy_t return almost undiminished at lags 12, 24 and 36. After both differences, $\Delta\Delta_{12} y_t$ keeps two clear autocorrelations, at lags 1 and 12: the signature of a moving average at lag 1 times a moving average at lag 12.

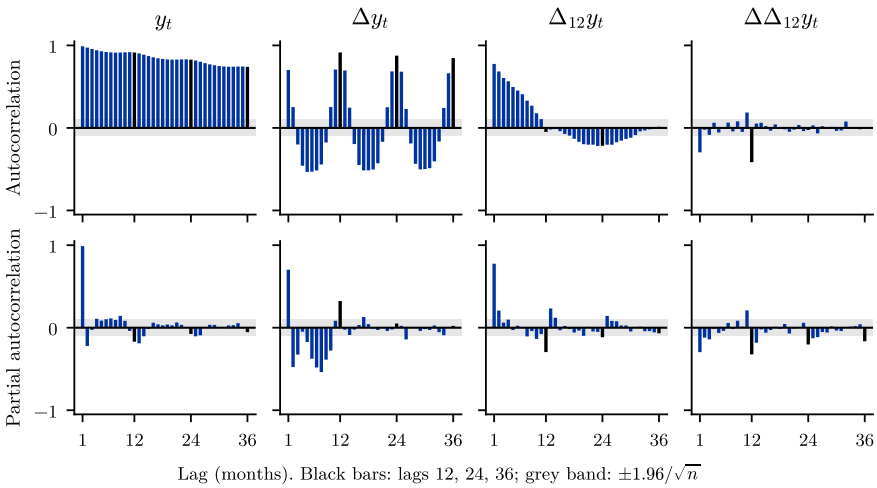


Figure 2. Sample autocorrelations (top) and partial autocorrelations (bottom), lags 1–36, January 1965 to December 1997.

BOX: WHY AIC CANNOT CHOOSE d

AIC $= -2 \ln \hat{L} + 2k$, with $\hat{L}$ the likelihood of the data the model describes. With $d = 1$ these are the Δy_t, with $d = 2$ the $\Delta^2 y_t$: different data, one observation fewer per difference.

Re-run, version 1's search finds its smallest AIC at $d = 1$ (best values 992.7, 845.6, 913.1 and 1066.1 for $d = 0, 1, 2, 3$). The comparison is not valid: d and D come from the tests and the correlogram; AIC then picks p, q, P, Q.

Selection. With $d = D = 1$ fixed, AIC compares 36 models ($p, q \le 2$; $P, Q \le 1$) on the same data. The smallest value, 187.41, belongs to the **airline model** of Box and Jenkins,

$$(1 - B)(1 - B^{12})\, y_t = (1 + \theta B)(1 + \Theta B^{12})\, \varepsilon_t, \tag{2}$$

with $\hat{\theta} = -0.392$ (s.e. 0.044), $\hat{\Theta} = -0.833$ (0.036) and $\hat{\sigma}^2 = 0.0906$. Its residuals pass the Ljung–Box test, $Q(m) = n(n+2) \sum_{k=1}^{m} \hat{\rho}_k^2/(n-k) \sim \chi^2_{m-2}$: $p = 0.70$, 0.77 and 0.90 for $m = 12, 24, 36$.

3. Forecasts for 1998–2001

Method. Each model is estimated once and forecast $h = 1, \ldots, 48$ months ahead from December 1997. Benchmarks: the seasonal naive forecast with drift, $\hat{y}_{T+h} = y_{T+h-12j} + j\bar{g}$ with $j = \lceil h/12 \rceil$ and $\bar{g}$ the mean twelve-month change, and additive Holt–Winters. For the ARIMA models the 95% interval is $\hat{y}_{T+h} \pm 1.96\, \hat{\sigma}\, (\sum_{i=0}^{h-1} \psi_i^2)^{1/2}$, with ψ_i the weights of the model's MA(∞) form.

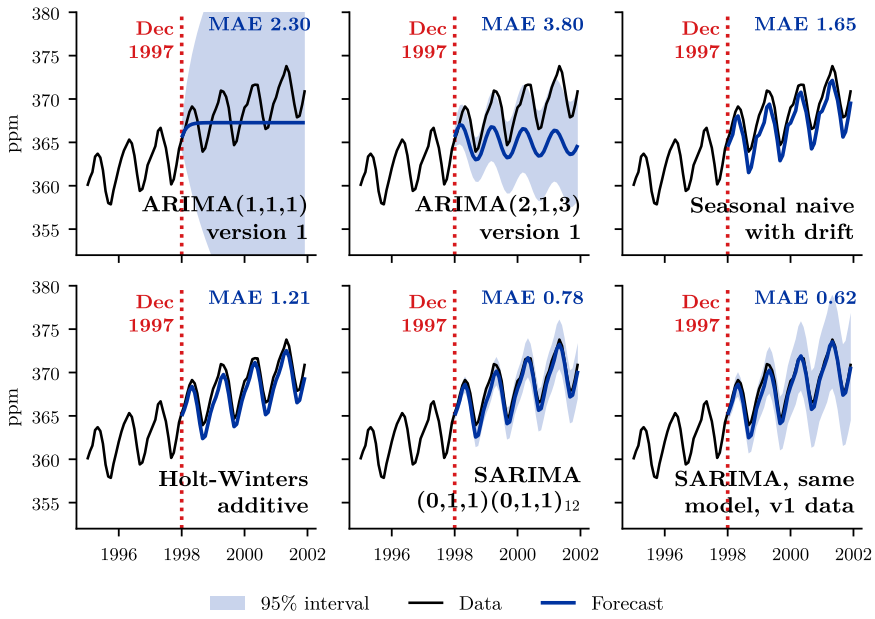


Figure 3. Forecasts made in December 1997, with the data from 1995. Panels 1, 2 and 6 use the data of version 1 (from 1958, filled months), the others the data of version 2 (from 1965).

Table 1. Forecast accuracy, January 1998 to December 2001 (48 months), ppm. Coverage: % of months inside the 95% interval.

Model	MAE	RMSE	Mean error	Coverage
ARIMA(1,1,1), v1	2.30	2.80	−1.57	100
ARIMA(2,1,3), v1	3.80	4.24	−3.77	69
Seasonal naive + drift	1.65	1.70	−1.65	–
Holt–Winters	1.21	1.25	−1.21	–
SARIMA $(0,1,1)(0,1,1)_{12}$	0.78	0.87	−0.78	85
same, v1 data	0.62	0.74	−0.60	100

Finding. Version 1's models cannot follow the data (Figure 3): ARIMA(1,1,1) flattens out within a few months and ARIMA(2,1,3) cycles around the level of late 1997 (its likelihood maximisation did not converge). In `statsmodels` an ARIMA with $d = 1$ has no constant, so these forecasts have no drift. The seasonal ARIMA has the smallest errors (Table 1), followed by Holt–Winters and the seasonal naive forecast. Every model forecasts too low (negative mean error): CO_2 grew faster in 1998–2001 than in the training years. The seasonal ARIMA's mean error is −0.94 ppm in 1998 and −1.01 in 1999, and 41 of 48 months fall inside its 95% interval.

Interpretation. The differences matter more than the choice among the remaining models. The same seasonal ARIMA estimated on version

1's data has a smaller error (0.62) but twice the innovation variance ($\hat{\sigma}^2 = 0.180$ against 0.091), an effect of the filled months: estimated from 1958 with the empty months skipped by the Kalman filter, $\hat{\sigma}^2 = 0.093$ and the error is 0.79. One forecast origin cannot rank these samples; it does separate seasonal from non-seasonal models.

4. Gold: is there an ARMA in the returns?

Why. Version 1 drew the price with "7-day" and "30-day" moving averages and a histogram of the price in pounds. The data are monthly and in US dollars, and a histogram of a trending price describes no distribution. The course question is whether the past of the price helps to forecast it.

Method. The monthly log return $r_t = 100\,(\ln P_t - \ln P_{t-1})$, and for it ARMA$(p, q)$ models $r_t = c + \sum_{i=1}^{p} \phi_i r_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$ with $p, q \le 2$. Forecasts are made one month ahead for the last 36 months (2023-10 to 2026-09), each from all earlier months, with the ARMA order chosen again by AIC every month. Benchmark: the historical mean $\bar{r}_t$; the out-of-sample $R^2 = 1 - \sum(r_t - \hat{r}_t)^2 / \sum(r_t - \bar{r}_t)^2$ is positive when a model beats it.

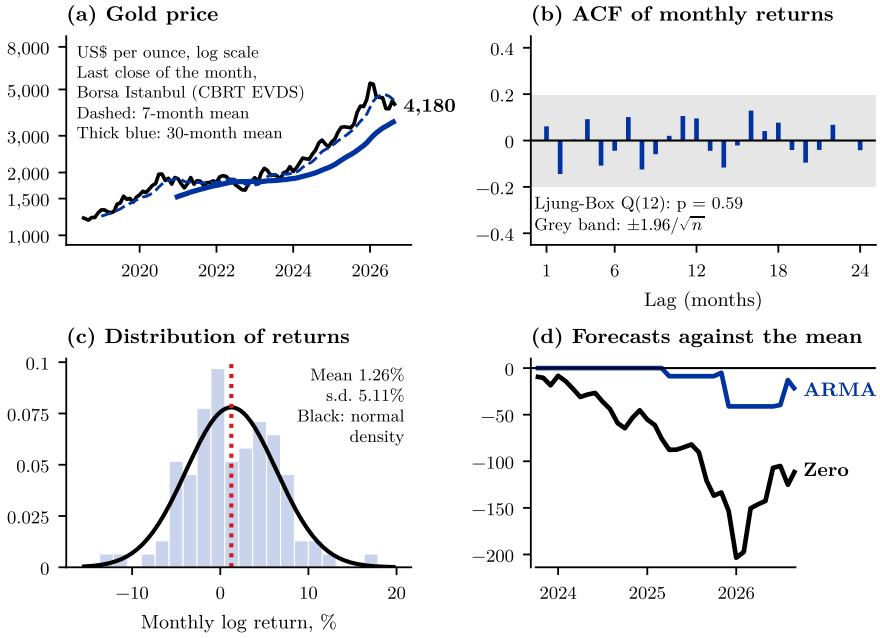


Figure 4. Gold, 2018-07 to 2026-09. (a) Price with version 1's smoothers, now labelled in months. (b) Autocorrelations of r_t. (c) Histogram of r_t and the normal density with the same mean and variance; red: the mean. (d) Cumulative squared error of the historical mean minus that of the model; a rising line means the model beats the mean.

Finding. The log price has a unit root (ADF $p = 0.78$), the return does not (ADF $p < 0.001$). The returns average 1.26% a month with a standard deviation of 5.11%, and their autocorrelations stay inside the band (Ljung–Box $Q(12)$, $p = 0.585$; Figure 4b). AIC prefers ARMA(2,2) by a small margin (598.8 against 601.0 for white noise), BIC prefers white noise. Out of sample the recursive AIC choice was white noise in all but 6 of 36 months, and its R^2 is -0.017; the random walk (zero forecast) has -0.077 (RMSE 6.32 and 6.51 against 6.27 for the mean).

Interpretation. Monthly gold returns behave like white noise around a positive mean: the Box–Jenkins steps end, correctly, with ARMA(0,0). The moving averages of version 1 describe the trend; they are not forecasts.

5. Conclusions and limitations

1. The order of differencing comes first: for CO_2 it is $d = D = 1$, and with it the simplest seasonal model, the airline model, forecasts four years ahead with an error below 1 ppm.
2. AIC is a tool for choosing among models of the same data; version 1's search over d violated that.
3. For gold the same steps end with white noise: past monthly returns do not improve on the historical mean.

Scope and verification note. Every number is written from `figures/*.csv` by `make_paper.py`; the unit tests check the computations against direct formulas, simulations with a known answer and the printed output of version 1. One forecast origin and four test years are a small sample. The gold series starts in July 2018, so version 1's 2012–2018 sample could not be reproduced.

References

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An AI coding assistant was used in writing the code and in preparing this note.